

Exploring Research and Tools in AI Security: A Systematic Mapping Study

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ABSTRACT With the pervasive integration of artificial intelligence (AI) in various facets of modern technology, the importance of AI security has been thrust into the spotlight. The field is rapidly evolving, with new challenges and solutions emerging at a swift pace. However, the breadth and depth of AI security research have not been comprehensively mapped in recent times, presenting a crucial need for an extensive review and synthesis of existing literature. Given the increasing reliance on AI in critical domains such as healthcare, finance, and national security, ensuring the resilience and trustworthiness of these systems is imperative. This survey fulfills the pressing need for a structured and comprehensive overview of the current research landscape, enabling researchers to address emerging threats and vulnerabilities effectively. This paper presents a systematic mapping study (SMS), aimed at identifying and classifying the prevailing research topics, tools, and frameworks in the field of AI security. A total of 123 studies were meticulously selected and analyzed, leading to the identification of key metrics, tools, standards, and research themes that are currently shaping the landscape of AI security research. This effort not only aids in distilling the collective wisdom of the research community but also sets a firm foundation for future work in this critical area. The findings from this SMS will serve as an invaluable guide for researchers and practitioners alike, enabling them to navigate the complexities of AI security and fostering the development of innovative, robust security solutions. This study also highlights significant gaps in the current literature, thereby outlining potential directions for new research initiatives.

INDEX TERMS AI Security, Generative AI Security, AI Risk Assessment, Detection and Defense, Ethical and Societal

I. INTRODUCTION

THE advent of the digital era has witnessed the meteoric rise of artificial intelligence (AI) as a cornerstone of innovation across numerous sectors. AI systems, characterized by their ability to learn, adapt, and execute complex tasks, are integral to advancing technology in today's world. However, as these systems become more sophisticated and embedded in critical infrastructure, the security of AI has emerged as a field of paramount importance and a considerable challenge [1] [2].

The need for secure AI systems transcends the traditional bounds of information security, venturing into realms where autonomous decision-making intersects with privacy, ethics, and reliability [3]. AI security is not merely about protecting systems against unauthorized access but ensuring these systems behave as intended even under malicious influences, thus preserving the integrity, confidentiality, and availability of critical digital assets [4] [5].

In recent years, the escalating complexity and frequency of AI-related attacks and security breaches have exposed the vulnerabilities in AI systems. These range from adversarial attacks disrupting machine learning models [6] to subtle manipulations aimed at AI-powered decision processes [7], leading to substantial financial and credibility losses for organizations. Addressing these vulnerabilities requires a concerted effort not only in the development of new AI models but also in the creation of resilient, secure applications.

This paper presents a systematic mapping study (SMS) on AI security, aiming to chart the existing security approaches and identify under-explored areas within the domain. The motivation behind this study stems from the observation that while numerous technical solutions have been proposed to address specific AI security issues, such as adversarial attacks and data poisoning, there is a notable lack of research that offers a holistic overview of AI security research and tools across its various facets [3] [8].

Moreover, the majority of AI systems are developed with a focus on functionality over security, often relegating the latter to a post-development afterthought [9]. This reactive approach to security in AI development processes poses significant risks and necessitates a shift towards proactive security integration. The SMS aims to characterize existing security methodologies, tools, and standards, offering a comprehensive survey that provides a structured guide to studies in the AI security field.

The rest of this systematic mapping paper is organized as follows. In Section II, we present the related work and compare our review with other reviews. Section III describes the systematic mapping process employed in this study. In Section V, we discuss the various AI security research topics identified in the mapping process. Section VI highlights the research gaps identified in this domain. Section VIII presents the results and discussion of our findings. Finally, in Section IX, we conclude by summarizing our key findings and outlining future research directions.

II. RELATED WORK

The rapid evolution of artificial intelligence (AI) has led to an influx of research on AI security, including reviews and surveys aimed at categorizing threats, analyzing defensive strategies, and exploring ethical implications. However, many of these works focus narrowly on specific topics, leaving gaps in comprehensively mapping the diverse and interconnected aspects of AI security. This section compares notable existing surveys with our systematic mapping study (SMS) to establish the distinct contributions of this work. Table 1 provides a detailed comparison of the works discussed.

Oseni, et al. [3] propose a framework for adversarial attack detection and mitigation, focusing on developing robust machine learning models that can withstand adversarial manipulation. The study also discusses advanced AI security techniques, including federated learning and reinforcement learning, to enhance model resilience. The paper provides a holistic cybersecurity perspective, covering both theoretical and practical aspects of AI security.

While the study provides valuable insights into adversarial attack strategies and countermeasures, it lacks a broader discussion on AI privacy, risk management, and regulatory compliance. The primary focus remains on adversarial threats, overlooking other significant security challenges such as data privacy breaches, model inversion risks, and AI governance frameworks.

M.Rahman et al. [10] present a systematic survey on AI security risks, offering a structured categorization of AI-specific attack vectors, including adversarial attacks, model inversion attacks, data poisoning, and membership inference attacks. The study highlights the impact of these attacks on AI integrity and privacy, detailing how adversaries exploit model weaknesses at both the training and inference stages. Moreover, the paper evaluates existing attack detection and mitigation techniques, such as adversarial training, differen-

tial privacy, homomorphic encryption, and secure multi-party computation.

Although the study provides an extensive taxonomy of AI threats, it is largely descriptive, offering a high-level overview rather than an in-depth analysis of attack behaviors and defense effectiveness. Additionally, while detection techniques are explored, the comparative efficacy of these methods remains unassessed, making it difficult to determine which strategies are most effective under real-world conditions. Another limitation is the lack of integration with industry security standards.

Gnitko et al. [8] work takes a regulatory and compliance-focused approach, offering a structured analysis of AI security standards across different jurisdictions. The study highlights how AI security frameworks vary between government policies, corporate security strategies, and international regulations, underscoring the fragmented nature of AI security governance. A key contribution of this study is the development of a taxonomy of AI security threats, categorizing risks based on their occurrence across the AI lifecycle (development, deployment, and operational phases). Additionally, the paper evaluates six major AI security frameworks, analyzing their applicability across machine learning models, deep learning applications, and AI-driven decision systems. By identifying gaps in existing security regulations, the study calls for a unified global standard to ensure consistent security practices across AI deployments.

Despite its strengths in regulatory analysis, the study takes a fragmented approach, focusing on policy implications rather than practical security solutions. While the importance of security compliance is well-articulated, the work does not explore how AI security standards can be effectively implemented in model development pipelines. Another shortcoming is the lack of empirical case studies or real-world evaluations, which would provide practical insights into how AI security regulations impact AI deployments in different industries.

Habbal et al. [11] introduce the AI Trust, Risk, and Security Management (AI TRiSM) framework, which provides a structured approach to managing AI security risks. The study explores how AI TRiSM can be integrated into AI governance, ensuring that AI-driven applications remain trustworthy, explainable, and resilient. The framework incorporates risk assessment methodologies, security compliance strategies, and AI reliability metrics. The study also highlights the growing concern of AI accountability, particularly in high-risk domains such as finance, healthcare, and law enforcement. By emphasizing transparency and risk-aware AI decision-making, AI TRiSM provides a governance model that aligns with ethical AI principles and regulatory compliance.

The AI TRiSM framework is conceptually well-defined, its technical implementation remains ambiguous. The study does not detail how AI TRiSM can be practically applied to AI security architectures. Furthermore, while the framework highlights AI risks, it does not account for real-world ad-

versarial threats, such as adaptive adversarial AI techniques. Another limitation is the absence of empirical validation, as the proposed framework lacks experiments or case studies that demonstrate its effectiveness in real-world AI security scenarios.

Xia et al. [12] present a mapping study on AI risk assessment, offering a structured classification of risk evaluation methodologies tailored for AI-driven systems. The study categorizes technical vulnerabilities, compliance risks, and ethical concerns, providing a taxonomy of AI security threats that can aid in designing risk-aware AI architectures. The study highlights the necessity of a connected AI risk assessment model, ensuring that AI security measures remain consistent, adaptive, and aligned with regulatory mandates. Additionally, the work explores threat simulation techniques, which enable organizations to evaluate AI models against adversarial manipulations, data poisoning, and model inversion attacks.

Despite its structured approach, the study remains largely theoretical, lacking practical validation through real-world security testing. Furthermore, while the study categorizes AI risks effectively, it does not provide an adaptive mechanism for updating AI risk models in response to evolving adversarial threats. Another limitation is the absence of detailed strategies for risk management in decentralized AI architectures, such as federated learning and edge AI systems.

Key Contributions of This Paper

This study stands apart by adopting a systematic mapping approach to chart the AI security research landscape holistically. The key differences include:

- i. **Comprehensive Scope:** Unlike prior works that focus on specific dimensions of AI security (e.g., adversarial attacks or risk management), this study categorizes research across diverse topics, including tools, frameworks, standards, threats, vulnerabilities, and lifecycle integration.
- ii. **Focus on Emerging Trends:** This SMS identifies recent advancements, such as generative AI security and large language model (LLM) vulnerabilities, areas that remain under-explored in previous reviews.
- iii. **Practical Relevance:** In addition to synthesizing academic research, this work emphasizes the practical applicability of findings for both researchers and practitioners, connecting methodologies to real-world AI development and deployment challenges.
- iv. **Identification of Gaps:** While existing reviews highlight challenges, this study systematically identifies and categorizes research gaps, offering a roadmap for future exploration in areas such as ethical AI security and multimodal AI systems.

By mapping the breadth and depth of AI security research, this work provides an integrative perspective that complements the more narrowly focused existing literature, bridging theoretical insights with practical implications.

III. SYSTEMATIC MAPPING PROCESS

The proposed SMS is a pivotal study for identifying and categorizing the breadth of research within the domain of AI security. It is meticulously designed to aggregate, synthesize, and analyze the existing body of knowledge, thereby uncovering trends, and gaps in the literature. The SMS (Figure 1) is conducted in accordance with guidelines established by Petersen et al. [13], ensuring a systematic and reproducible approach to the review process.

1) Phase 1: Research questions and search strategy

We initiate the review process by outlining the scope and research questions that will steer our investigation, followed by establishing a search strategy.

Research questions specification - Our SMS is structured around specific research questions, which guide the search for relevant literature. Four distinct research questions, along with their rationale, have been proposed and are presented in Table 4. These questions were formulated to direct the mapping study towards yielding insights that are both comprehensive and pertinent to the field of AI security.

Developing search strategy - The search strategy involves identifying relevant literature using well-defined keywords and query strings across multiple databases

2) Phase 2: Extracting keywords and collecting relevant studies

In this phase, relevant keywords are identified from the research questions to facilitate an effective search strategy, and studies are subsequently collected based on these defined keywords.

Keywords and Query Strings - We begin with an initial list of keywords, including AI/ML security, AI/ML risk assessment, AI/ML vulnerabilities, adversarial attacks, and large language model security (Table 5). This list is then expanded by incorporating additional keywords found in the selected references.

Databases - The following electronic databases are searched to ensure comprehensive coverage:

- IEEE Xplore
- ACM Digital Library
- SpringerLink
- ScienceDirect
- Google Scholar
- Scopus
- Web of Science

3) Phase 3: Selection and classification

To ensure that only relevant and high-quality studies are included in the mapping study, we establish clear inclusion and exclusion criteria. The inclusion and exclusion criteria ensure that only relevant and high-quality studies are included in the mapping study.

Inclusion Criteria:

- Studies published in peer-reviewed journals and conferences (Table 2 and Table 3)
- Studies focusing on AI security methods and applications.

TABLE 1. Summary of AI Security Surveys and Reviews

Surveys/Reviews	Reference	Main Contribution	Limitation of the Work
Security and Privacy for Artificial Intelligence: Opportunities and Challenges	[3]	- Proposed frameworks for adversarial attack and defense. - Discussed advanced techniques like federated learning.	Focused only on adversarial threats, lacking coverage of broader security challenges like privacy and risk management.
Security Risk and Attacks in AI: A Survey of Security and Privacy	[10]	- Categorized AI attacks (e.g., adversarial, poisoning, inversion). - Discussed detection techniques and defenses like adversarial training.	Narrow focus on attacks and defenses without integrating tools or standards.
Systematic Overview of AI Security Standards	[8]	- Developed a taxonomy of AI security threats. - Analyzed six key AI security frameworks and their applicability.	Fragmented focus on specific standards with limited integration.
Artificial Intelligence Trust, Risk, and Security Management (AI TRiSM): Frameworks, Applications, Challenges, and Future Research Directions	[11]	- Introduced the AI TRiSM framework for trust, risk, and security management. - Explored challenges and applications in specific domains like finance and health-care.	Lacked systematic categorization of research themes and broader security solutions.
Towards Concrete and Connected AI Risk Assessment (C2AIRA): A Systematic Mapping Study	[12]	- Explored and categorized AI risk assessment methodologies. - Provided actionable insights for risk-aware AI system design.	The study primarily concentrates on AI risk assessment strategies, thereby overlooking other critical aspects of AI security.
Exploring Research and Tools in AI Security: A Systematic Mapping Study	This Study	- Integrates multiple dimensions of AI security, including tools, frameworks, standards, threats, and lifecycle integration. - Identifies emerging trends like generative AI security and LLM vulnerabilities. - Identifies and evaluates practical security tools and frameworks. - Systematically identifies and categorizes research gaps for future exploration.	

- Latest Industrials reports and findings.
- Studies published in English.

Exclusion Criteria:

- Non-English publications.
- Studies without sufficient technical details, empirical results, or clear theoretical contributions (e.g studies without rigorous analysis or formal verification insights).

Classify Studies by Categories:

Once the relevant papers are selected, we classify them into key thematic categories within AI Security based on keywords and core topics to ensure each paper is mapped to the most relevant category (see Figure 2 and Table 5). The first category, *Threat Models in AI*, includes research on taxonomies of AI threats, adversarial models, attack surfaces of machine learning models, and AI system vulnerabilities across different attack vectors. *AI Risk Assessment* focuses on risk quantification methodologies, case studies on AI-driven cyber risks, AI reliability evaluations, and governance models addressing security concerns. Studies in *Detection and Defense Techniques* propose or evaluate adversarial attack detection methods, robustness enhancement strategies, AI model verification, and defenses against adversarial and poisoning attacks. *Generative AI Security* covers security challenges in LLMs and diffusion models, adversarial attacks on generative models, backdoor vulnerabilities, and model manipulation techniques. The *Ethical and Societal Implications of AI Security* category addresses AI bias and fairness, ethical concerns surrounding adversarial AI attacks, and the

responsible use of AI in offensive security. *AI Security Tools and Frameworks* includes research on open-source security tools, frameworks for securing AI applications, and standardization efforts in AI security.

4) Phase 4: Quality assessment:

After defining the research questions, extracting keywords, and categorizing the selected studies into relevant themes, we proceeded to analyze the quality of these studies. This quality assessment was conducted based on various criteria, including:

- **Credibility of the source:** Evaluating the trustworthiness of the publication and the reputation of the authors.
- **Relevance to the research questions:** Ensuring the studies directly address the specific questions and themes identified.
- **Peer review status:** Checking whether the study has undergone peer review, a process where experts critically evaluate its methodology, findings, and overall contribution to the research area.
- **Citation count:** Reviewing the number of times the study has been cited by other researchers as a measure of its impact and recognition within the academic community.
- **Depth of analysis:** Evaluating the comprehensiveness and depth of the study's analysis and discussion.

IV. RESEARCH MOTIVATIONS AND QUESTIONS

The explosive growth of the applications over the last few years has been paralleled by a corresponding intensification

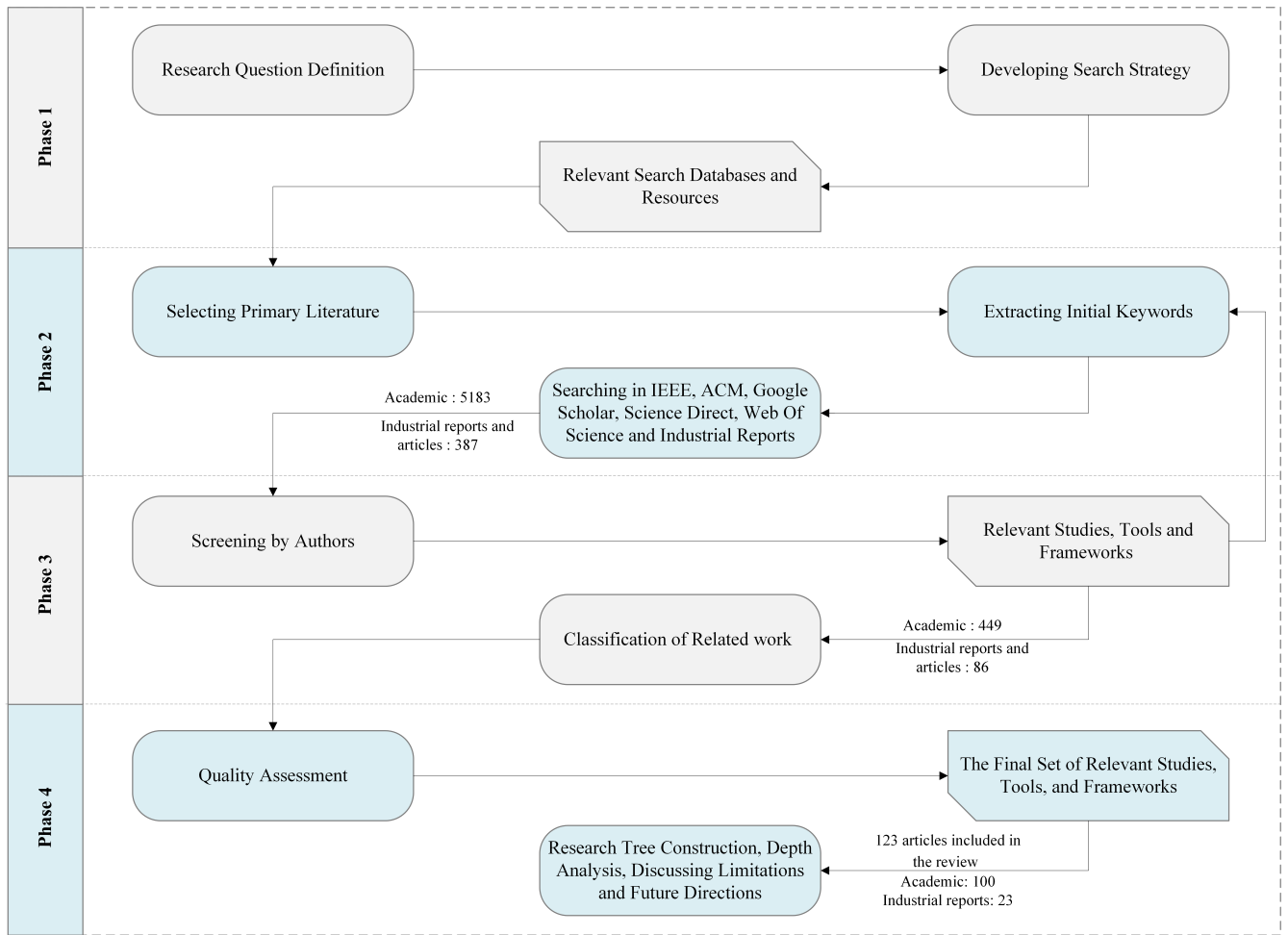


FIGURE 1. Steps of the SMS conducted

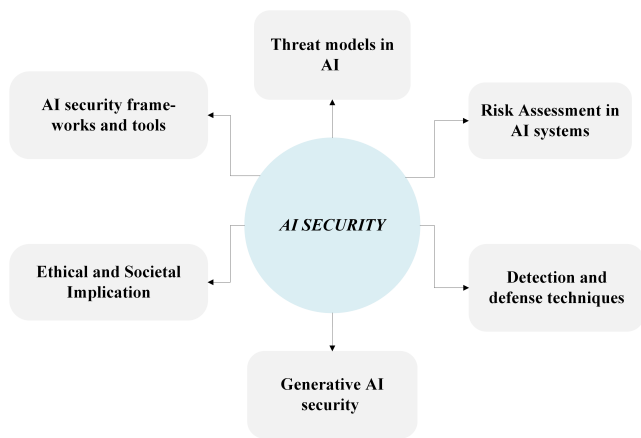


FIGURE 2. AI Security Research papers domains.

TABLE 2. Number of Articles Obtained from Various Journals

Journals	Number of Articles
IEEE Access	7
ACM Computing Surveys	6
Expert Systems with Applications	3
IEEE Transactions on Dependable and Secure Computing	2
Springer Journals	5
Elsevier Journals	7
Journal of Information Warfare	1
Journal of Computer and Communications	1
IEEE Wireless Communications	1
International Journal of Artificial Intelligence and Machine Learning	1
International Journal of Security and Privacy in Pervasive Computing (IJSPPC)	1
American Physical Society (APS)	1
Nature Machine Intelligence	1
IEEE Transactions on Software Engineering	1

in the complexity and sophistication of security threats, particularly in the realm of AI. With AI systems becoming ubiquitous, the discipline of Secure AI Engineering has gained prominence, advocating the integration of security consid-

erations to proactively withstand attacks and maintain the fundamental principles of security: confidentiality, integrity, and availability.

Recent advancements in AI have brought forth an array

TABLE 3. Number of Articles Obtained from Conferences

Conferences	Number of Articles
ACM SIGSAC Conference on Computer and Communications Security	3
IEEE/CVF International Conference on Computer Vision	2
NeurIPS	2
AAAI/ACM Conference on AI, Ethics, and Society	3
ACL 2024	2
Empirical Methods in Natural Language Processing(EMNLP)	1
International Conference on Learning Representations (ICLR)	2
International Conference on Machine Learning 2024	1
Annual Reliability and Maintainability Symposium (RAMS)	1
IEEE/ACM International Conference on AI Engineering	1
NDSS 2024	1
New Security Paradigms Workshop (NSPW)	1
ACM Conference on Fairness, Accountability, and Transparency	1
SAFECOMP 2022	1
HSCC 2021	1
AAMAS	1
IJCAI	1
AI Safety 2022	1
International Conference on Agents and Artificial Intelligence	1
39th Annual Computer Security Applications Conference (ACSAC)	1
2021 IEEE International Conference on Cyber Security and Resilience (CSR)	1
2021 IEEE International Conference on Mobile Networks and Wireless Communications	1
2020 12th International Conference on Cyber Conflict (CyCon)	1
28th International Conference on Evaluation and Assessment in Software Engineering	1
IEEE Symposium on Security and Privacy (SP)	1
2019 IEEE/ACM 41st International Conference on Software Engineering (SEIP)	1
2017 ACM on Asia Conference on Computer and Communications Security (Asia CCS)	1

of technical innovations, yet the incorporation of security within these systems remains a multifaceted challenge. The majority of AI systems are engineered with a primary focus on functionality, relegating security to a secondary concern often addressed post-development, a strategy that is increasingly untenable in today's security-conscious environment [14] [15]. Moreover, conventional software development processes are often inadequate for addressing the unique security requirements of AI systems, leading to vulnerabilities that can be exploited through a myriad of attack vectors, from adversarial manipulation to systemic flaws [16] [17].

This literature review is motivated by the pressing need to systematically map the existing body of knowledge in AI security. By evaluating how the current literature addresses the security of AI systems, this study aims to reveal prevailing research trends, identify effective security measures, and discern gaps in the literature where future research efforts could be most beneficial. Furthermore, this study endeavors

to augment the collective understanding of AI security measures and practices, providing a robust foundation for the development of secure AI applications.

In alignment with these motivations, the study poses the following research questions (RQs):

RQ1) What is the publication frequency of AI security research?

This question helps determine the level of attention and priority given to this topic by the research community over time. This analysis covers the period from 2014 to 2025, highlighting trends and significant milestones in AI security research. The number of publications analyzed for each year is shown in Fig 3., which includes until the end of January 2025.

To facilitate a deeper examination, we divided the study period into two distinct time periods:

2014-2018: Initial Growth Phase- In the early years of this study, AI security started to gain traction as a vital field of research. The rise of machine learning and AI technologies brought about new security challenges. During this period, researchers focused on foundational aspects of AI security, such as securing machine learning models, detecting adversarial attacks, and developing initial frameworks for AI system security.

2019-2024: Rapid Expansion and Diversification- From 2019 onwards, the publication rate of AI security research increased significantly. There is significant growth in the number of studies, driven by several factors, including advancements in AI technologies that required robust security measures, high-profile cyberattacks leveraging AI, increased funding for AI security research, etc.

This research question not only focuses on identifying the number of publications each year but also delves into understanding the underlying factors influencing publication trends. For instance, the analysis considers how emerging threats in AI security, such as adversarial attacks, model poisoning, and backdoor vulnerabilities, have contributed to spikes in research interest over the years. In recent years, a significant shift in focus is observed towards securing Generative AI (Gen AI) systems, driven by the rapid adoption of models like ChatGPT, DALL-E, and other large language and generative models. This shift reflects the increasing concerns over vulnerabilities specific to Gen AI, such as adversarial prompts, data leakage, and misuse in generating deepfakes or malicious content. By examining these evolving trends, the study highlights how the research community has responded to real-world challenges and technological advancements, particularly the rise of Gen AI, shaping the current landscape of AI security research.

RQ2) What are the emerging sub-fields within AI security?

The field of AI security is diverse and rapidly growing, with new sub-fields of its own arising to meet the demands of integrating AI technology into modern systems. AI security has branched into distinct areas that address the specific risks, defenses for the AI systems. This question emphasizes vari-

TABLE 4. Research Questions on AI Security

Research Question	Rationale
What is the publication frequency of AI security research?	By answering this question, researchers will understand the growth rate and attention devoted to AI security research over time.
What are the emerging sub-fields within AI security?	This question helps identify new research frontiers and the expanding scope of AI security.
What are the threat models in AI systems and how can they be systematically mitigated across the AI lifecycle?	Answering this question provides insight into the prevalence of AI threats and vulnerabilities, helping organizations understand the various attack vectors that AI systems are exposed to, as well as the necessary defenses required to safeguard these systems throughout their lifecycle.
How is AI security risk assessed across different industries?	With this question, researchers can compare risk assessment approaches and challenges in sectors like finance, healthcare, and defense.
What tools and frameworks have been developed for AI security?	This question aims to explore the landscape of tools and frameworks designed for securing AI systems, including those used for red teaming, adversarial testing, threat detection, and vulnerability assessment.

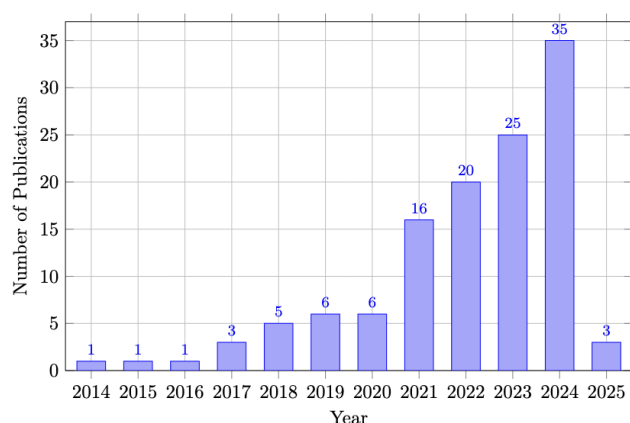


FIGURE 3. Number of publications analysed in AI security for each year

ous focal points, including adversarial machine learning, Gen AI security, large language model (LLM) security, ethical AI, detection and defense techniques.

To provide a clearer picture of the landscape, in this paper highlights the most prominent research topics currently gaining traction within AI security. These topics encapsulate the challenges and innovations shaping the field and offer insight into the directions in which research is heading. Section V provides a detailed exploration of each sub-field, offering a comprehensive understanding of the tools, frameworks, and strategies that are driving the future of AI security.

RQ3) What are the threat models in AI systems and how can they be systematically mitigated across the AI lifecycle?

The frequency of security incidents has been rising steadily over the past years, emphasizing the escalating challenges faced by organizations in protecting AI systems [18] [19]. This trend points to an increasing sophistication among threat actors and a corresponding need for advanced defensive strategies. The AI systems are vulnerable to a range of threats, including evasion attacks, poisoning attacks, backdoor attacks, and model replication [20]. These vulnerabilities can be exploited at various stages of the AI lifecycle, from data collection to deployment, and require a multi-level approach to defense [21] [22]. Specific vulnerabili-

ties, such as adversarial attacks and poisoning attacks, are particularly concerning and require careful consideration in the design and implementation of AI systems. The need for a comprehensive approach to AI security, considering the entire system lifecycle and its integration with larger IT infrastructures, is emphasized [23]. As detailed in Section V (A), the threat models in AI are multifaceted. They range from poisoning attacks, where adversaries manipulate training data, to backdoor attacks that embed malicious triggers in the system's training data, leading to specific misbehaviors upon activation. Evasion attacks, which involve crafting adversarial inputs to fool models during inference, and model stealing, where attackers replicate models through queries, are also prevalent. Furthermore, threats like membership inference and attribute inference, which exploit the generalization ability of AI models to extract sensitive data, further underscore the need for robust security measures. The development of autonomous threat hunting and the identification of risks from machine-generated text, such as misinformation or harmful content, are additional avenues explored in AI security. Understanding these threat models is fundamental to building resilient AI systems capable of withstanding a wide range of adversarial actions, as highlighted in Section V (A).

RQ4) How is AI security risk assessed across different industries?

The proliferation of AI technologies also brings significant security risks that need to be assessed and managed meticulously [24] [25]. Different industries face unique challenges and vulnerabilities in AI security, making it crucial to understand these risks for the development of effective mitigation strategies. In Section V (B), we explore the vulnerabilities of AI systems across various industries, such as financial services, healthcare, manufacturing, autonomous vehicles, and retail, by identifying potential hazards, analyzing their likelihood and impact, and discussing mitigation strategies like robust AI systems, regular audits, and human oversight to enhance AI security and safeguard against emerging threats.

These risks require tailored security frameworks, such as regular audits, adversarial testing, data encryption, and robust training approaches. The adoption of collaborative defense

mechanisms and continuous monitoring is essential across industries to safeguard AI systems and ensure their integrity, as highlighted by industry standards and security protocols. Mitigation strategies focus on enhancing the resilience of AI systems by proactively identifying weaknesses through rigorous testing, incorporating human oversight, and adopting ethical guidelines to prevent biases and vulnerabilities. Regular updates to risk management protocols and the integration of advanced encryption and access controls further help in reducing the impact of potential security breaches. Moreover, fostering collaboration across industries allows for shared insights and resources to strengthen defenses against evolving AI security threats.

In particular, to address these security risks, industries must adopt comprehensive AI security frameworks. These include:

Regular Audits and Monitoring: Continuous auditing and real-time monitoring of AI systems are crucial to detecting and mitigating potential security breaches. Regular security testing helps ensure that vulnerabilities are identified early and addressed proactively, which is emphasized by multiple AI risk management frameworks like Microsoft's and Google's security protocols [26] [27].

Adversarial Testing: Implementing adversarial testing, such as through adversarial robustness frameworks, allows organizations to simulate attacks that AI models might face. This process helps identify weaknesses in AI systems before they can be exploited by malicious actors [28].

Data Encryption and Access Controls: Ensuring data security is critical in AI applications. Data must be encrypted both in transit and at rest, with strict access controls that limit access to authorized personnel only. This principle is often emphasized in industry standards like the NIST AI Risk Management Framework [29] and the OWASP AI Security guidelines [30].

Data sanitization: it ensures that the input data is clean, accurate, and free from malicious manipulations that could otherwise compromise the model's integrity and performance.

Robust training: robust training approaches help mitigate attacks by attempting to develop a training algorithm which limits the effectiveness of malicious points by design rather than removing these points altogether [28].

Bias and Fairness Assessments: Regularly assessing AI models for biases and ensuring that they operate fairly and transparently.

Collaborative Defense Mechanisms: Collaboration across industries, sharing knowledge and resources to combat AI security threats, strengthens defenses. Initiatives like Microsoft's Counterfit [31] tool and collaborative frameworks from MITRE emphasize the importance of collective efforts to enhance AI security.

RQ5) What tools and frameworks have been developed for AI security?

The development of AI security tools and frameworks is crucial for safeguarding AI systems from a wide range of adversarial attacks and vulnerabilities. AI security frameworks

are designed to defend machine learning models against malicious inputs aimed at manipulating or misleading them. Among the most common types of attacks are evasion, where small, often imperceptible changes to input data cause a model to make incorrect predictions, and data poisoning, where attackers corrupt the training data, leading to degraded model performance. These frameworks focus on enhancing model resilience, enabling them to function reliably even under adversarial conditions. This involves robust training techniques, model verification processes, and continuous monitoring, all aimed at maintaining the integrity and stability of AI systems.

In addition to these defensive measures, AI red teaming tools are designed to simulate variety of adversarial behaviors, including evasion attacks, data poisoning, backdoor attacks and stress-test models, allowing organizations to detect vulnerabilities before they are exploited. By leveraging red teaming tools, organizations can assess how well their AI systems respond to threats, evaluate the model's robustness to secure their AI systems during both the development and deployment phases.

On the other side, AI blue teaming tools focus on continuously safeguarding AI systems through real-time threat monitoring, anomaly detection, and defensive mechanisms. These tools are essential for building a robust defense posture, ensuring that AI models are able to detect and mitigate attacks as they occur.

To provide a more detailed examination of the tools and frameworks developed to address these challenges, refer to Section V (F). Table 7 and Table 8 outline the key frameworks and tools, offering a comprehensive view of how both classical machine learning (ML) and generative AI (GenAI) systems are safeguarded across various scenarios.

V. AI SECURITY

A. THREAT MODELS IN AI

Threat models in AI play a crucial role in identifying and mitigating potential risks associated with AI systems [33]–[43], [118]–[123]. These models help researchers and practitioners understand the various ways AI systems can be attacked and how to defend against these threats. Here are some of the key threat models in AI, based on recent research:

Poisoning Attacks: These occur during the training phase of machine learning (ML) models. An attacker introduces malicious data into the training set, causing the model to learn incorrect patterns. This can degrade the model's performance or manipulate its behavior for specific inputs. Defenses against poisoning attacks include data sanitization and robust training methods. **Backdoor Attacks:** Similar to poisoning, backdoor attacks involve embedding hidden triggers in the training data that cause the model to behave maliciously when these triggers are activated. This type of attack is particularly challenging to detect and defend against because the model appears to perform normally on typical inputs [124].

TABLE 5. Primary Studies Topics and Keywords

AI Security Research Topics	References	Keywords
Threat models in AI	[32]–[44]	Adversarial attacks, Backdoor attacks, Evasion attacks, Model poisoning, etc.
Risk Assessment in AI systems	[18]–[21], [23], [45]–[51]	Vulnerability analysis, AI reliability, Security assessment, Risk Mitigation etc.
Detection and defense techniques	[52]–[67]	Adversarial defense, Mitigation techniques, Robust AI, etc.
Generative AI Security	[68]–[99]	Prompt injection, Jailbreaking, Hallucinations, LLM robustness, etc.
Ethical and Societal Implication	[100]–[106]	Fairness, AI ethics, Accountability, Transparency, etc.
AI security frameworks and tools	[26], [29], [31], [107]–[117]	Security frameworks, Testing tools, Security audit, etc.

Evasion Attacks: These occur at inference time, where adversaries craft inputs (adversarial examples) that cause the model to make incorrect predictions. Techniques to generate adversarial examples range from simple gradient-based methods to more sophisticated approaches. Defenses include adversarial training and input preprocessing [124].

Quantum-based adversarial attacks: Quantum adversarial machine learning (QAML) introduces a new frontier in AI security, where quantum computing can both amplify the robustness and vulnerabilities of machine learning models [125] [126] [127] [128] [129]. As quantum computing advances, traditional security strategies in machine learning need to evolve to address quantum-enabled threats. Quantum adversaries may employ quantum algorithms to bypass current defenses, leveraging quantum properties such as superposition and entanglement to manipulate or degrade the performance of machine learning models. These quantum adversarial threats range from simple noise insertion at the lowest level to more sophisticated attacks, such as quantum cryptanalysis and the development of advanced quantum algorithms aimed at undermining model integrity [126] [130]. Defending against such attacks calls for the integration of quantum-secure design principles, including post-quantum cryptography, quantum-resistant neural networks, and transparency in development and deployment. These efforts aim to ensure the safety and trustworthiness of machine learning systems in a quantum-powered world [126]. Furthermore, the rapid development of quantum processors and algorithms necessitates proactive measures like quantum data anonymization and continuous monitoring of quantum systems to mitigate the evolving risks posed by quantum adversaries.

Model Stealing: In model stealing attacks, an adversary attempts to replicate the functionality of a target model by querying it and using the responses to train their own model. This can lead to intellectual property theft and potential misuse of the model. Defenses include limiting the information returned by the model and monitoring for unusual querying patterns [124].

Membership Inference and Attribute Inference: These attacks aim to determine whether specific data points were part of the training set (membership inference) or to infer sensitive attributes of the data used to train the model. These attacks exploit the way models generalize from training data, posing privacy risks. Defenses involve techniques like differential privacy and regularization methods [7].

Autonomous Threat Hunting: This involves using AI to proactively search for vulnerabilities and threats in systems, leveraging AI's ability to analyze vast amounts of data and identify patterns. Autonomous threat hunting aims to enhance traditional cybersecurity measures and is pivotal for dealing with sophisticated cyber threats [131].

Threats from Machine-Generated Text: As natural language generation (NLG) models become more advanced, they can be exploited to generate harmful content, spread misinformation, or deceive users. Threat models for NLG focus on understanding the potential adversaries, their capabilities, and objectives, and developing detection and mitigation strategies to prevent misuse [132].

These threat models are essential for developing robust AI systems that can withstand adversarial actions

B. AI RISK ASSESSMENT: DETAILED ANALYSIS

As the deployment of AI systems becomes increasingly widespread across various industries, assessing the associated risks has become a crucial aspect of ensuring the secure and responsible implementation of these technologies. AI systems, while offering significant benefits, also introduce a range of potential hazards that can have serious consequences. These risks are not only technical but can extend to ethical, operational, and strategic areas. The assessment of AI risks aims to identify, analyze, and mitigate the potential dangers posed by AI technologies, ensuring their safe integration into critical sectors such as healthcare, finance, manufacturing, and transportation. Without a comprehensive risk assessment framework, organizations may be exposed to vulnerabilities that could lead to breaches, exploitation, and even societal harm [45]–[51]. This section examines the various hazards that AI systems may encounter, the techniques created to evaluate such risks, and the effective mitigation techniques put in place (Figure 4)

i) AI security risk in different industries

Different industries face unique challenges and vulnerabilities when it comes to AI security, and understanding these can help in developing robust mitigation strategies. The motivation to focus on the following industries when assessing AI security risks stems from the crucial role AI

plays in these sectors and the magnitude of risk associated with AI-related vulnerabilities and security breaches.

Financial Services

In the financial services industry, AI is widely used for fraud detection, algorithmic trading, risk management, and customer service automation. The security risks here include data breaches, algorithmic biases, and the manipulation of AI systems. A significant concern is adversarial attacks where malicious entities can exploit vulnerabilities in AI models to execute fraudulent transactions or manipulate financial markets. For instance, the study by Xie et al. [133] highlights the potential for adversarial attacks to manipulate stock prices.

Healthcare

AI in healthcare is revolutionizing diagnostics, personalized medicine, and predictive analysis [134]. AI and ML algorithms are capable of processing large volumes of healthcare data, enhancing precision in areas such as medical imaging and genomics. However, the security risks in this high-stakes domain are profound. One major risk is adversarial attacks, where malicious actors introduce subtle changes to input data, such as medical images, leading AI systems to produce incorrect outcomes, potentially causing misdiagnosis and inappropriate treatment plans [135] [136]. Moreover, vulnerabilities in peripheral devices and communication channels expand the threat surface.

To mitigate these risks, regular audits and encryption protocols are crucial to ensure the integrity of the systems. Additionally, incorporating Human-in-the-Loop (HiTL) frameworks can alleviate the risks associated with overreliance on AI by maintaining human oversight in critical decision-making processes. HiTL systems allow clinicians to verify and, if necessary, override AI-generated diagnoses, ensuring that final decisions reflect both human expertise and AI precision [137]. This balance ensures that AI remains a powerful tool for enhancing healthcare while minimizing security vulnerabilities and errors.

Manufacturing

In the manufacturing industry, AI optimizes production lines, predicts maintenance needs, and enhances supply chain management. The primary security risk here involves industrial espionage and sabotage. AI systems can be targeted to disrupt production processes or steal proprietary information. Securing AI systems against cyber-physical attacks is crucial for maintaining operational integrity and protecting intellectual property.

Autonomous Vehicles

The automotive industry, particularly with the advent of autonomous vehicles (AV), faces unique AI security challenges. Autonomous systems rely heavily on AI for navigation, obstacle detection, and decision-making. A critical risk is physical adversarial attacks on these systems, which can cause vehicles to misinterpret signals or obstacles. A widely studied example is the manipulation of STOP signs with adversarial patches, where subtle changes to the appearance of the sign can cause the vehicle's AI system to misclassify or fail to detect the sign entirely, potentially leading to dangerous

outcomes [138]. Research on securing AI in AVs emphasizes the need for robust defense mechanisms and improving the certifiable robustness of AI models.

Retail

In retail, AI is used for customer behavior analysis, inventory management, and personalized marketing. Security risks include data breaches and the manipulation of AI systems for fraudulent activities. For example, AI algorithms that analyze customer data to predict purchasing behavior can be targeted to steal consumer information or manipulate purchasing trends. A report by Deloitte [139] highlights the need for robust AI security frameworks to protect consumer data and ensure the integrity of AI-driven marketing strategies.

ii) AI System Risk Factors

AI systems are vulnerable to a wide range of threats, from hostile external attacks to technical malfunctions. These risks can be broadly categorized as follows: strategic risks pertain to decisions made using AI that may have unfavorable effects; ethical risks are related to the potential harm AI systems may cause to human values and rights; operational risks are related to the functioning and performance of AI systems; and technical risks are related to vulnerabilities [140]. To effectively address these risks, it is crucial to expand AI risk assessment and mitigation strategies. One approach is to prioritize the development of robust AI systems that are resilient to external attacks and technical malfunctions. This can be achieved through rigorous testing and validation procedures to identify and rectify vulnerabilities in AI systems. Additionally, proactive measures should be taken to ensure that AI systems adhere to ethical guidelines and do not infringe upon human rights and values. Moreover, establishing comprehensive operational protocols can help mitigate operational risks associated with the functioning and performance of AI systems. By addressing these different categories of risks, organizations and researchers can work towards enhancing the safety and reliability of AI technologies. One crucial aspect of expanding AI risk assessment and mitigation strategies is to consider the potential impact of AI decisions on various aspects of society. This involves not only evaluating the immediate effects of AI systems, but also anticipating and preparing for potential long-term consequences. Ethical considerations, such as ensuring fairness, transparency, and accountability in AI decision-making processes, are essential components of comprehensive risk assessment and mitigation strategies.

iii) Risk Assessment Methodologies

Risk assessment in AI involves identifying potential hazards, analyzing the likelihood of these hazards occurring, and evaluating the potential impact. Popular frameworks for AI risk assessment as illustrated in Table 7 offer structured approaches to assess risks systematically and recommend measures to mitigate them. One of the key aspects of risk assessment is the identification of potential hazards. This involves examining the specific functions and components of AI systems to pinpoint areas of vulnerability. Once potential hazards are identified, the next step is to analyze

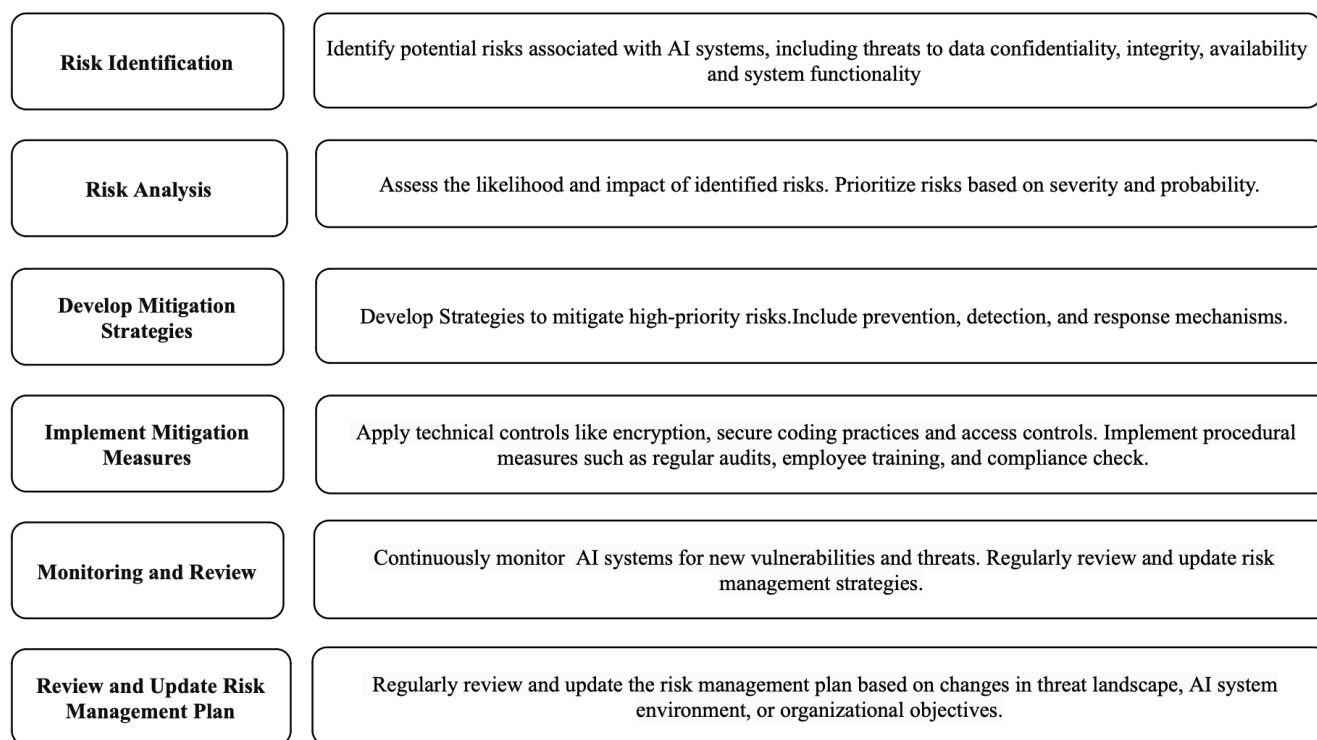


FIGURE 4. AI Security Risk Assessment (adapted from [141], [142] [143]).

the likelihood of these hazards occurring. Factors such as system complexity, data inputs, and external dependencies play a crucial role in determining the probability of various risks. Furthermore, evaluating the potential impact of these hazards is essential for understanding the potential consequences of AI-related risks. This involves considering the potential harm to individuals, organizations, or society as a whole. By assessing the likelihood and potential impact of risks, stakeholders can prioritize their mitigation efforts effectively. In addition to these methodologies, continuous monitoring and adaptation of risk assessment strategies are crucial. AI systems and their operating environments are constantly evolving, requiring ongoing assessments to stay ahead of emerging risks. This proactive approach allows for the timely identification and mitigation of new threats, ultimately enhancing the overall resilience of AI systems

C. DETECTION AND DEFENSE TECHNIQUES

Vulnerability detection in AI systems is critical for preventing exploitation and ensuring the integrity and confidentiality of AI-driven operations. This subsection delves into how vulnerabilities are identified, assessed, and managed in the realm of artificial intelligence, emphasizing the tools and techniques used to detect and respond to vulnerabilities effectively [52]–[64], [144].

i) Importance of Vulnerability Detection

The detection of vulnerabilities within AI systems is crucial for several reasons:

Preventing Attacks: Early detection allows for the pre-

emptive addressing of security flaws that could be exploited by attackers. **Maintaining Trust:** Ensuring the security of AI systems is vital for maintaining user trust in technology applications, especially in sensitive areas like healthcare and finance. **Regulatory Compliance:** Many industries are subject to stringent regulatory requirements concerning data security, making vulnerability detection a compliance necessity. [145] [146]

ii) Techniques for Vulnerability Detection

Vulnerability detection in AI systems often employs a combination of automated tools and expert analysis, including:

Static Analysis: Examining the AI's codebase without executing the program to find vulnerabilities that could lead to security breaches. **Dynamic Analysis:** Running the AI systems in controlled environments to monitor for unexpected behaviors or outputs that indicate security weaknesses. **Fuzz Testing:** Introducing malformed or unexpected inputs to the system to identify potential vulnerabilities that occur under unusual or unexpected conditions. **Penetration Testing:** Simulating cyber attacks on the AI systems to assess their robustness and the effectiveness of existing security measures.

Challenges in AI Vulnerability Detection:

Detecting vulnerabilities in AI systems poses unique challenges, including - **Complexity of AI Models** The complexity and often opaque nature of machine learning algorithms can make it difficult to identify where vulnerabilities may exist. **Evolving Threat Landscape:** As AI technology advances, so do the tactics of attackers, requiring continuous updates to

vulnerability detection methodologies. Scalability: With AI being implemented at a large scale across different sectors, ensuring comprehensive vulnerability coverage remains a logistical and technical challenge.

iii) Mitigation Strategies

Once risks are assessed, appropriate mitigation strategies must be employed. These include technical measures such as implementing robust training protocols to reduce the susceptibility to adversarial attacks, applying anomaly detection systems to catch unusual behaviors indicative of security breaches, and designing AI systems with inherent security features like encryption and access controls.

Another critical aspect is the establishment of robust incident response protocols. Organizations should develop comprehensive procedures for identifying, containing, and resolving AI system failures or security breaches with dedicated response teams and regular drills.

By integrating robust risk assessment methodologies, proactive mitigation strategies, and a commitment to ethical best practices, organizations and researchers can work towards enhancing the safety, reliability, and societal impact of AI technologies [147].

D. GENERATIVE AI SECURITY

Generative AI (GenAI) has emerged as a transformative technology with significant implications across numerous domains, but with its rise come substantial security challenges that differentiate it from traditional machine learning models. Unlike conventional ML models, which operate with a narrower set of tasks and structured goals, Generative AI models such as Large Language Models (LLMs) and diffusion models are characterized by their autonomy in generating diverse, high-quality content. This capability, while transformative, significantly broadens the potential attack surface, leading to new security vulnerabilities [148] [149].

Large Language Models (LLMs) such as GPT-4, Gemini, and other transformer-based models have revolutionized natural language processing. However, their advanced capabilities also introduce significant security challenges. Here is a comprehensive overview of the detection and defense techniques specifically aimed at securing LLMs [68]–[79].

i) Security Risks in LLMs

Adversarial Attacks: Adversarial attacks are one of the most prominent security risks in LLMs, where attackers craft inputs to deceive the model into generating harmful or incorrect outputs. These attacks can trick the model into producing biased, offensive, or even malicious content by subtly altering the input prompts [150].

Studies have shown that LLMs like ChatGPT can be manipulated into bypassing their ethical filters through carefully crafted input prompts [151]. Such attacks exploit vulnerabilities in how LLMs interpret language, allowing adversaries to cause misbehavior in the models, such as generating phishing emails or inappropriate content. This issue is particularly relevant in conversational applications, where LLMs are expected to interact with users and produce reliable, ethical

responses. The absence of human oversight in many cases amplifies the risk, making it a critical area of concern for ensuring the safe deployment of conversational systems.

ii) Data Poisoning and Backdoor Attacks

Data poisoning involves the manipulation of training data to introduce harmful behaviors into a model. This type of attack can occur during both pre-training and fine-tuning phases, but the effectiveness and mechanisms can differ significantly depending on when the attack is executed.

During Pre-training: When data poisoning occurs at the pre-training stage, it can subtly influence the model's learning process on a foundational level. Given the vast datasets typically used for pre-training large language models (LLMs), such attacks can be difficult to detect but may have more generalized effects, potentially impacting many downstream applications. However, the attacker's control over specific outputs may be less precise, as the poisoned data is diluted among a massive amount of benign data, making it harder to ensure predictable malicious behavior.

During Fine-tuning: Fine-tuning, on the other hand, usually involves smaller, more domain-specific datasets, making it a more focused point of attack. Poisoning during fine-tuning can be more effective for introducing specific backdoors because the attacker has greater control over the data influencing the model at this stage. Fine-tuned models are often used for specialized tasks, and introducing backdoors at this point can allow attackers to precisely trigger malicious behaviors or generate harmful outputs when specific inputs are provided.

Backdoor attacks could be introduced during the training or fine-tuning of LLMs, causing models to behave maliciously when triggered by specific inputs [152]. These attacks can cause LLMs to behave unpredictably or even compromise security by revealing sensitive information or generating harmful outputs.

iii) Model Stealing

Attackers can replicate the model's functionality by querying it extensively and using the responses to train their own model. Rate limiting, query monitoring, and deploying techniques like watermarking can help in detecting and preventing model stealing [153].

iv) Jailbreak Attacks

Jailbreak attacks occur when adversaries bypass built-in safety mechanisms by crafting input prompts that cause the model to produce restricted or harmful outputs.

The sophisticated prompts can bypass LLM safety filters, causing models to generate offensive or dangerous content, even in cases where ethical guidelines should prevent such behavior [152].

These attacks pose a serious threat, particularly in automated systems, where LLMs could be tricked into performing illegal or harmful actions without proper oversight.

Indirect Malicious Instruction Execution:

Cross-Site Scripting (XSS) and CSRF: Attackers can embed malicious instructions in web content that LLMs access.

Securing the interfaces and implementing strict content validation protocols are essential defense measures [154].

Detection Techniques Anomaly Detection: Monitoring model inputs and outputs for unusual patterns that deviate from normal behavior. This includes tracking the frequency and nature of queries to detect potential extraction or injection attempts [150].

Behavioral Analysis: Utilizing AI techniques to analyze the model's behavior over time and detect deviations that might indicate ongoing attacks. This involves creating behavioral baselines and continuously comparing current behavior against these baselines [150].

v) Adversarial Training

Enhancing the model's robustness by training it on both benign and adversarial examples. This makes the model more resilient to adversarial inputs during real-world use (Generative Models for Security: Attacks, Defenses, and Opportunities, 2022).

vi) Rate Limiting and Throttling

Limiting the number of queries a user can make in a given time period to prevent model stealing and DoS attacks. This involves setting thresholds and employing rate-limiting mechanisms [155].

vii) Watermarking and Fingerprinting

Embedding unique identifiers in the model's responses to detect unauthorized use. Watermarking helps trace back any stolen model or its outputs to the original source [151].

viii) Robust Interface Design

Ensuring that interfaces through which the LLM interacts with external systems are secure. This includes validating inputs and outputs and implementing security protocols to prevent injection attacks [156]. Addressing these require a multi-faceted approach combining detection, defense, and continuous monitoring. As these models become more integrated into various applications, addressing their security vulnerabilities becomes increasingly critical to prevent misuse and protect sensitive information. Continuous research and development of new techniques are essential to stay ahead of emerging threats in the AI landscape.

E. ETHICAL AND SOCIETAL IMPLICATIONS OF AI SECURITY

AI security encompasses a broad range of ethical and societal issues that need to be carefully addressed to ensure the development of responsible and trustworthy AI systems [100]–[104]. Here are some of the key ethical and societal implications of AI security:

i) Bias and Fairness

One of the most significant ethical concerns in AI is bias. AI systems can inadvertently learn and propagate biases present in their training data, leading to unfair and discriminatory outcomes. This is particularly problematic in sensitive applications such as hiring, law enforcement, and healthcare. Efforts to mitigate bias involve developing and implementing fairness metrics, bias detection tools, and adopting fair training practices. Researchers emphasize the importance of

fairness in AI and advocate for the development of algorithms that can ensure equitable treatment across different demographic groups [157].

ii) Accountability

Determining accountability in AI systems is complex, particularly when these systems operate autonomously. There is a need for clear guidelines and frameworks to assign responsibility when AI systems fail or cause harm. This includes developing legal and regulatory frameworks that define the roles and responsibilities of AI developers, users, and other stakeholders [158].

iii) Security and Misuse

AI systems are susceptible to various security threats, including adversarial attacks, data poisoning, and model theft. These threats can compromise the integrity and reliability of AI systems, leading to harmful consequences. It is essential to implement robust security measures to protect AI systems from malicious actors. Additionally, there is a risk of AI being used for malicious purposes, such as creating deepfakes or automating cyberattacks, which poses significant societal risks [159].

iv) Societal Impact

AI has the potential to significantly impact society by changing the nature of work, influencing social interactions, and shaping economic structures. There is a need to consider the broader societal implications of AI, including its impact on employment, social equity, and access to resources. Policymakers and researchers are urged to adopt a holistic approach to AI development, ensuring that the benefits of AI are distributed equitably across society and that potential negative impacts are mitigated [157].

F. AI SECURITY TOOLS AND FRAMEWORKS

AI security frameworks provide structured approaches to securing AI systems, ensuring they operate safely, reliably, and in compliance with legal and ethical standards. These frameworks help organizations assess risks, implement controls, and monitor AI systems throughout their lifecycle.

AI security tools are essential for protecting AI systems from various threats, such as adversarial attacks, data breaches, and model theft. These tools help detect, prevent, and mitigate security risks, ensuring the integrity and reliability of AI applications.

Our exploration unearthed a diverse arsenal of tools and techniques tailored to fortify AI against the plethora of threats. Among these, machine learning models designed to detect and counter adversarial attacks have shown a notable surge in research interest. Tools like adversarial training frameworks and robustness assessment platforms are being continuously refined to address the evolving complexity of threats that AI systems face. These frameworks, summarized in Table 7, highlight the diverse approaches to AI governance and security, catering to various stages of AI lifecycle management and different regional or sector-specific needs. And the tools, detailed in Table 8, represent the cutting-edge technologies being used to secure AI systems. They highlight

TABLE 6. Taxonomies on AI Security

Titles	References	Limitations
Artificial intelligence security threat, crime, and forensics: Taxonomy and open issues	[160]	Traditional forensic techniques may not solve AI-related crimes effectively. Challenges in establishing novel strategies for AI forensics.
Towards AI safety: A taxonomy for AI system evaluation	[142]	Primarily theoretical with limited empirical validation
Adversarial machine learning: A taxonomy and terminology of attacks and mitigations	[161]	Theoretical limitations on adversarial robustness in AI systems. Challenges in designing secure machine learning algorithms for critical task
Backdoor attacks and defenses in federated learning: State-of-the-art, taxonomy, and future directions	[162]	Focused specifically on federated learning, not generalizable to all AI systems
A Security Risk Taxonomy for Large Language Models	[151]	Does not cover all potential misuses or attack vectors. Encourages future research to uncover evolving threats.

the ongoing shift from reactive defense mechanisms to predictive security tools, leveraging AI's predictive capabilities to preempt potential breaches [154] [163].

In this context, the integration of explainability tools has been identified as a critical factor in AI security. The ability to interpret AI decision-making processes is not only essential for trust and transparency but is increasingly recognized as a cornerstone of security, providing insights into potential vulnerabilities within AI systems.

G. LIMITATIONS OF AI SECURITY TAXONOMIES

The analyzed taxonomies (Table 6) each address unique security and evaluation challenges, crafted to respond to specific operational and threat contexts in AI. However, each taxonomy also has limitations that reflect challenges in its scope, applicability, or empirical support.

AI System Evaluation Framework: This taxonomy provides a comprehensive, lifecycle-wide approach to AI system assessment, addressing both component and system levels for Narrow and General AI. By mapping safety evaluations to stages across the development, deployment, and operational lifecycle, it emphasizes stakeholder accountability and long-term risk management. This holistic structure supports standardized evaluation and accountability across diverse AI environments and stakeholder roles, fostering a resilient approach to AI safety that encompasses both model-centric and system-wide concerns.

Limitation: The taxonomy is primarily theoretical with limited empirical validation, which may restrict its practical applicability. The reliance on theoretical constructs could be refined with empirical studies to ensure its effectiveness in real-world scenarios.

Prompt-Based LLM Security Taxonomy: Uniquely organized around user, model, and third-party targets, this taxonomy focuses on vulnerabilities inherent in prompt-based interactions within large language models (LLMs). Structured along the Confidentiality, Integrity, and Availability (CIA) triad, it categorizes prompt manipulation risks and prescribes defenses that are modular and interaction-focused. The emphasis on targeted defenses aligns with LLM-specific threats where prompt integrity is critical to secure user-

model interactions, enabling a nuanced approach to mitigating manipulation risks in LLM deployments. **Limitation:** This taxonomy does not encompass all potential misuse cases or attack vectors, focusing primarily on known threats. As LLMs evolve, new risks are likely to emerge, suggesting a need for ongoing updates to address future vulnerabilities comprehensively.

AI Threats, Crime, and Forensics Taxonomy: Structured through the lens of criminology, this taxonomy examines AI's dual role as both a tool and a target of criminal activity. By categorizing AI threats along this dual-use framework, it supports forensic analysis, legal accountability, and regulatory oversight. This approach is particularly suited to applications where AI's societal impact must be carefully managed, enabling systematic tracking of AI's involvement in criminal activity and offering structured guidance for ethical compliance and regulatory adherence in AI-driven environments. **Limitation:** Traditional forensic techniques may not effectively address AI-related crimes, indicating challenges in adapting established investigative methods to AI-specific contexts. Developing specialized forensic approaches for AI crimes is necessary to fully address these challenges.

Federated Learning Backdoor Taxonomy: This taxonomy addresses the specific vulnerabilities in federated learning (FL) systems, focusing on backdoor threats introduced through data poisoning and model poisoning attacks. It emphasizes anomaly detection and robust training as key defenses, adapted for Federated Learning's decentralized, privacy-preserving framework. Its compatibility with secure aggregation and differential privacy highlights its alignment with FL's non-centralized, privacy-first architecture, allowing for effective backdoor mitigation without compromising FL's core design principles. **Limitation:** The taxonomy is highly specialized for federated learning and is not generalizable to all AI systems. Its narrow focus limits its applicability outside federated learning environments, suggesting a need for expanded frameworks that adapt these security mechanisms to other decentralized or centralized AI systems.

Common Themes and Key Differences: Across these taxonomies, several shared themes emerge, reflecting foundational principles in AI security and evaluation:

Shared Emphasis on Security and Risk Mitigation: All taxonomies underscore the importance of rigorous security and risk assessment. Each taxonomy incorporates robust defenses against unauthorized access, data manipulation, and system integrity breaches, using established principles like the CIA triad or lifecycle-based risk assessment to ensure resilient AI operations.

Component-Level vs. System-Level Evaluation: Each taxonomy distinguishes between component-specific evaluations (e.g., model accuracy, data quality) and broader system-level assessments. This layered approach ensures that both the core AI model and the broader deployment context are systematically evaluated for security vulnerabilities and performance.

Ethics and Accountability: Ethical considerations, regulatory compliance, and stakeholder accountability are woven into each taxonomy. The AI System Evaluation Framework emphasizes accountability through lifecycle mapping, while the AI Threats taxonomy specifically addresses societal impacts through a forensic, legal-oriented perspective. Meanwhile, the LLM taxonomy emphasizes ethical deployment by mitigating interaction-based risks, and the Federated Learning taxonomy promotes privacy preservation and security in a decentralized context.

Key differences arise in the taxonomies specific scopes and methodologies, tailored to their application contexts. The AI System Evaluation Framework provides lifecycle-wide, standardized assessments suitable for diverse applications, while the LLM and Federated Learning taxonomies focus narrowly on interaction-specific and decentralized security risks, respectively. The AI Threats taxonomy is unique in its forensic orientation, aligning with law enforcement and regulatory needs in contexts where AI impacts are directly linked to societal and legal outcomes.

VI. RESEARCH GAPS

Despite the advancements, our study identifies critical gaps in current AI security research.

One such gap is the need for standardized benchmarks for AI security, which would enable the consistent evaluation of AI security tools and methodologies. Standardization would help unify the assessment of various approaches, allowing for reliable comparisons and establishing best practices across the industry. Such benchmarks could also facilitate the deployment of robust security measures in real-world applications where interoperability and compliance are essential.

Another significant gap is the lack of research on security implications in emerging domains, such as autonomous systems and the Internet of Things (IoT), where AI's interaction with the physical environment introduces new security risks. In these areas, threats can extend beyond data integrity and privacy concerns, posing risks to physical safety, infrastructure stability, and public trust in technology. Developing security frameworks that consider both digital and physical interactions in these domains is crucial.

Moreover, while there is extensive literature on the security of Generative AI (GenAI), especially concerning large

language models (LLMs), most of this research highlights the vulnerabilities and challenges rather than proposing concrete solutions. As these technologies integrate further into our daily lives, the focus must shift towards hardening these systems against attacks. There is a pressing need for targeted research on the defense of AI assistants, including LLMs, as well as multimodal generative AI models, which combine text, image, and audio generation capabilities. These systems are increasingly used in sensitive applications, and their complexity creates multifaceted security challenges that require new approaches to threat detection, adversarial defense, and model transparency.

While Quantum Adversarial Machine Learning (QAML) opens up new avenues for enhancing AI systems, it simultaneously exposes critical research gaps in defending against quantum-based adversarial attacks. Current security mechanisms lack the sophistication to counteract quantum-enabled threats that exploit unique quantum properties like superposition and entanglement to bypass traditional defenses. The absence of standardized frameworks for quantum-resistant neural networks and the limited exploration of post-quantum cryptographic methods leave AI systems vulnerable to emerging quantum adversarial strategies, such as quantum cryptanalysis and advanced algorithmic manipulations. Additionally, there is a pressing need for adaptive defense mechanisms, such as real-time quantum system monitoring and quantum data anonymization, to address the dynamic nature of quantum threats. The interplay between quantum advancements and AI vulnerabilities highlights a significant gap in ensuring the robustness, trustworthiness, and safety of machine learning models in the face of quantum-enabled adversarial challenges.

Additionally, as AI security research evolves, the exploration of ethical and governance frameworks remains underdeveloped. Given the ethical implications of deploying AI in critical areas like healthcare, finance, and law enforcement, there is a need for more detailed guidelines on responsible and transparent AI security practices. This includes developing frameworks that address issues such as bias, accountability, and transparency to ensure AI systems uphold societal values while maintaining robust security.

Addressing these identified gaps by developing solutions-focused research in GenAI security, enhancing frameworks for emerging AI applications, and establishing ethical governance standards will be essential to advancing the field of AI security. These efforts will support the responsible and resilient adoption of AI technologies across industries.

VII. AI SECURITY TOOLS LIMITATIONS

In addition to gaps in AI security research, there are several limitations in the development and deployment of AI security tools that hinder their effectiveness and adoption.

Adaptability to Evolving Threats: One of the core challenges is that AI security tools struggle to keep pace with the rapid evolution of adversarial tactics. As AI models become more sophisticated, so do the methods of attack, from

TABLE 7. AI Security Frameworks

Frameworks	Developer/Publisher	Description	Key Components	Limitations
AI Risk Management Framework (AIRMF) [29]	National Institute of Standards and Technology (NIST)	Focuses on managing risks for AI systems across various stages, leveraging a comprehensive set of risk principles	All principles, all stages, region-agnostic, sector-agnostic	Lack of standardization, complexity, resource-intensive, limited adaptability, and voluntary adoption.
AI Ethics Guidelines [107]	European Union (EU)	Offers a comprehensive framework for ethical and trustworthy AI, including risk assessment and mitigation.	Human Agency and Oversight, Robustness and Security, Privacy and Data Governance	Non-binding, vague principles, difficult for small organizations, slow adaptation, and limited global alignment.
Responsible AI Framework [26]	Microsoft	A framework for developing and deploying AI systems responsibly, including risk assessment and mitigation.	Fairness, Reliability and Safety, Privacy and Security, Inclusiveness, Transparency, Accountability	Proprietary focus, limited transparency, resource-intensive, generalized guidance, and voluntary adoption.
Model AI Governance Framework [108]	Personal Data Protection Commission (Singapore)	Guides organizations in implementing sound AI governance practices.	Descriptive, all stages, region-agnostic, sector-agnostic	Non-mandatory, resource-intensive, limited global applicability, ambiguous principles, and slow adaptation to evolving risks.
MITRE ATLAS [164]	MITRE	Community knowledge base for adversarial AI threats	Focuses on generative AI vulnerabilities, provides real-world case studies and tactics.	Specialized focus on cybersecurity, complexity, requires expertise, resource-heavy, limited scalability, and static approach to new threats.
NSW Artificial Intelligence Assurance Framework [109]	Government of New South Wales (Australia)	Assures the responsible deployment and operation of AI technologies.	Reusable anywhere with adjustments, all stages	Non-binding, resource-intensive, ambiguous principles, limited global relevance, and slow adaptation to AI advancements.
Secure AI Framework (SAIF) [27]	Google	Provides security-by-default principles to protect AI systems, focusing on threats specific to AI.	Strong security foundations, Detection and response, Model theft protection, Prompt injection protection, Data integrity safeguards	Proprietary focus, limited transparency, resource-intensive, generalized approach, and voluntary adoption.
Algorithm Impact Assessment Tool (AIA) [110]	Government of Canada	Designed for assessing the impact of algorithms, particularly in the public sector, with a focus on mitigation risks.	Planning and requirements analysis, design, testing	Voluntary use, public sector focus, resource-intensive, lacks detailed technical guidance, and scalability challenges for smaller organizations.
Fundamental Rights and Algorithm Impact Assessment (FRAIA) [111]	Ministry of the Interior and Kingdom Relations, Netherlands	Evaluates AI applications against fundamental rights to ensure compliance and ethical deployment.	Public sector focus, all stages	Public sector focus only, limited global applicability, lacks technical depth, and difficult to scale for smaller projects.
Ethics and Algorithms Toolkit [112]	City and County of San Francisco, Harvard Data-Smart, and Data Community DC	Helps public agencies deploy AI systems responsibly.	Public sector focus, all stages	U.S.-centric focus, voluntary framework, resource-heavy for small agencies, lacks technical depth, and struggles with rapidly evolving AI contexts.
Databricks AI Security Framework (DASF) [165]	Databricks AI	Designed to help organizations securely build, deploy, and manage AI models, focusing on data protection, model governance, and compliance.	Data protection, model risk management, compliance adherence, and secure deployment for AI and ML workflows.	Platform-specific, resource-intensive, limited transparency, generalized guidance, and slow adaptation to emerging AI risks.

TABLE 8. AI Security Tools

Tool Name	Developed By	Category	Main Functionality	Notable Features	Weaknesses
PyRIT [113]	Microsoft	Open-Source	Red teaming automation for generative AI systems	Automates adversarial red teaming, supports risk detection in content generation and fairness.	Requires substantial prompt engineering, lacks formal reporting, and has relatively lower attack success rates for single-step attacks [166].
Adversarial Robustness Toolbox (ART) [114]	IBM & Linux AI & Data Foundation	Open-Source	Defense against adversarial AI threats	Supports various models and data types, defends against common adversarial threats.	Limited focus on natural language vulnerabilities, requires domain expertise to configure effectively.
Counterfit [31]	Microsoft	Open-Source	Security testing automation tool for AI systems	Environment and model agnostic, supports penetration testing and red teaming.	Limited ability to handle complex and evolving adversarial scenarios due to static configurations.
IBM Security Verify	IBM	Commercial	Identity access management using AI	Enhances consumer and workforce protections, offers a cloud-native approach.	Narrow focus on access management, not designed for comprehensive red teaming of LLMs.
Garak [167]	NVIDIA	Open-Source	Vulnerability scanning of large language models	Focuses on vulnerabilities in LLMs, including hallucinations and harmful language outputs.	Relies heavily on static attack datasets, has high margin of error (MOE), and lacks robust customizability [166].
Guardrails [168]	Guardrails AI	Open-Source	Quantitative privacy assessments of ML models	Identifies data leakage risks, provides privacy scores and detailed reports.	Limited coverage of broader vulnerability types, focuses mainly on privacy-related aspects.
AI Fairness 360 [169]	IBM	Open-Source	An open-source toolkit for detecting, measuring, and mitigating bias in machine learning models.	Provides over 70 fairness metrics, multiple bias mitigation algorithms for pre-, in-, and post-processing, and compatibility with popular ML frameworks like scikit-learn and TensorFlow.	Bias-focused, lacks adaptability for general-purpose security testing or adversarial robustness.
PurpleLlama [170]	Meta	Open-Source	Provides tools for cybersecurity evaluations and input/output safeguards for generative AI systems	Enables testing of insecure code generation, filtering of unsafe content, and collaboration between red and blue teams for risk mitigation.	Focuses on code generation vulnerabilities, less suitable for other LLM vulnerability categories like jailbreak or multi-turn attacks.
Prompt Fuzzer [171]	Prompt Security	Open-Source	Tests and strengthens GenAI system prompts against dynamic AI-based threats	Simulates over a dozen types of GenAI attacks, is interactive and user-friendly, supports 20 LLM providers, and continuously integrates new attack simulations.	Heavily relies on pre-defined attack scenarios, limiting adaptability to novel vulnerabilities.
Rebuff [115]	Protect AI	Commercial	Defense against prompt injection attacks	Offers multi-layered defense for AI applications from prompt injection (PI) attacks.	Narrowly focused on prompt injection, not designed for detecting broader vulnerability
ModelScan [116]	Protect AI	Commercial	Scans documents and Jupyter notebooks for vulnerabilities	Evaluates for PII, open source code issues, and more, particularly useful for AI development environments.	Limited scope for dynamic adversarial testing, focuses primarily on static vulnerabilities in code and documents.

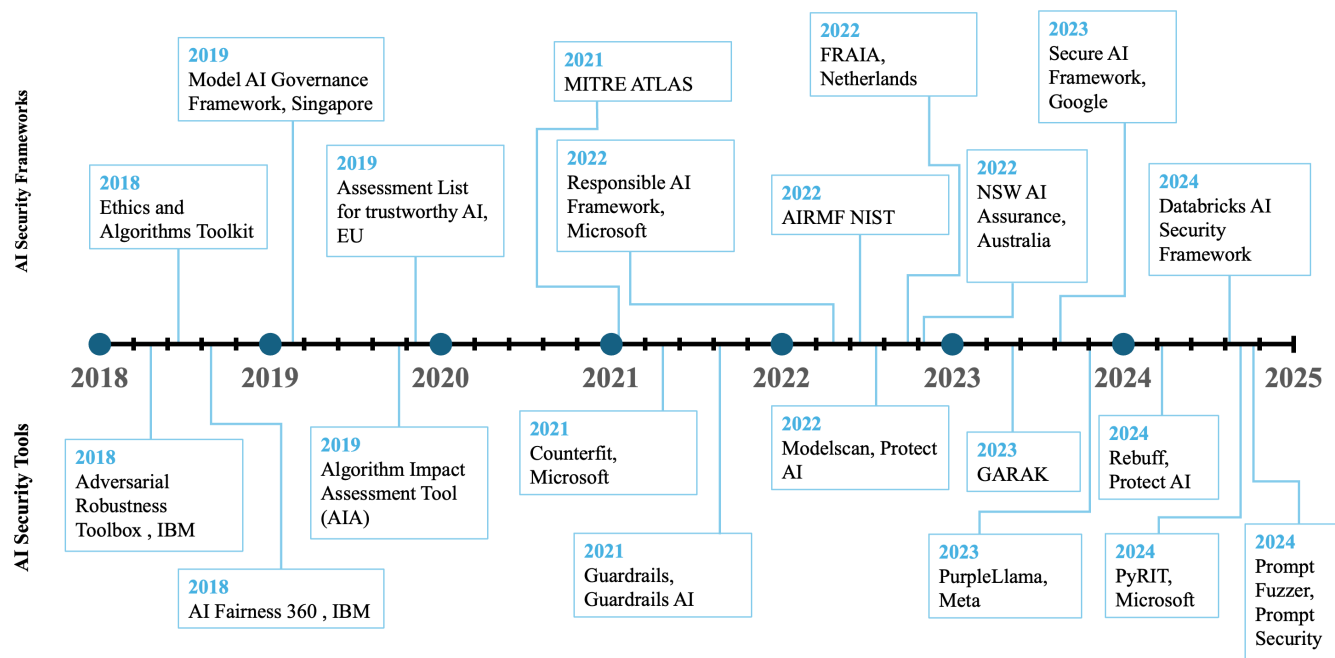


FIGURE 5. AI Security Tools and Frameworks Timeline.

subtle data poisoning to complex adversarial manipulations. Security tools must be highly adaptable to detect and counter new and unforeseen threats effectively, which demands continuous updates and agile development.

Lack of Scalability Across Models and Applications: Many AI security tools are tailored to specific types of AI models or applications, making it difficult to apply them across different platforms or domains. This lack of scalability limits their usability, especially in large, complex systems that utilize diverse AI components. Tools that work with one type of model, such as computer vision, may not function effectively with natural language models, highlighting a need for more versatile solutions.

Performance Overhead and Computational Cost: Effective AI security tools often introduce significant computational overhead, which can impact the efficiency of AI systems. Techniques like adversarial training, real-time threat monitoring, and robust model verification can slow down the system's performance, making these security tools impractical for time-sensitive applications like autonomous vehicles or high-frequency trading. Reducing the computational load without sacrificing security remains a key challenge.

VIII. RESULTS AND DISCUSSION

The systematic mapping study pivots, merging quantitative data with qualitative insights to present a cohesive narrative of AI security research findings. Through the lens of meticulously collated studies, this segment delineates the prevailing research topics, methodological approaches, and prevalent tools that underpin AI security, while also venturing into the less charted territories that represent future frontiers. However, while academic research lays an essential foundation, it

is crucial to consider how these methodologies translate into real-world industry practices, which often adapt and refine academic models to meet practical, scalable needs.

Synthesis of Secure AI Engineering Practices: The mapping study also synthesized secure AI engineering practices from the selected literature. A trend towards embedding security considerations into the AI design and development process, rather than as a post-deployment afterthought, was apparent. This proactive approach indicates a paradigm shift in how security is perceived in the context of AI. Security-by-design principles are being advocated, integrating threat modeling, risk assessment, and security testing early in the AI development lifecycle [172].

Thematic Analysis of AI Security Research: Thematic analysis revealed that research in AI security is becoming increasingly granular, with specialized themes such as privacy-preserving AI, secure AI in edge computing, and blockchain for AI security gaining momentum. These themes highlight the necessity for security measures that are not only technologically adept but are also congruent with legal and ethical standards.

Transparency and Explainability: AI security tools often operate as black boxes, making it difficult for users to understand how they identify threats or vulnerabilities. This lack of transparency can hinder trust and limit the tools' applicability in sectors where explainability is critical, such as healthcare or finance. Tools that integrate transparency and explainability, providing insight into the decisions made during security assessments, would be more broadly useful and trusted across regulated industries.

Limited Focus on Multimodal and Generative AI Models: While there are security tools available for traditional

machine learning models, few are designed for complex, multimodal models (those combining text, image, and audio) or generative AI systems, such as large language models. These models have unique vulnerabilities that require specialized security measures. For instance, large language models are vulnerable to prompt injection attacks and jailbreaking, yet few tools address these specific threats effectively. Enhancing AI security tools to cover multimodal and generative AI models is essential as these systems become more integrated into everyday applications.

Challenges in Real-World Integration and Testing: Security tools are often developed and tested in controlled environments, but their efficacy can diminish when deployed in real-world settings. Factors such as data variability, user behavior, and unanticipated system interactions introduce complexities that are difficult to replicate in testing phases. As a result, AI security tools may not perform as expected outside laboratory conditions, underscoring the need for more robust testing methods that mimic real-world conditions.

Addressing these challenges is vital for advancing the reliability and applicability of AI security tools. By improving scalability, transparency, adaptability, and efficiency, AI security tools can become more practical and effective in safeguarding diverse AI systems in real-world environments.

IX. CONCLUSION

As we conclude this systematic mapping study on AI security, several key insights emerge, offering a comprehensive portrayal of the current research landscape while also shedding light on vital areas for future exploration. This study has synthesized diverse perspectives and findings from a wide array of scholarly works, enabling a deeper understanding of AI security's multifaceted nature. Firstly, the current state of AI security research, as reflected in the papers analyzed, demonstrates significant advancements in identifying and mitigating various threats and vulnerabilities associated with AI systems. These developments highlight the continued need for vigilance and innovation in this area, especially as AI technologies become more integrated into different aspects of society. Through a comprehensive gap analysis, this study reveals several under-explored areas that warrant further research. These gaps include the need for more sophisticated adversarial defense mechanisms, holistic risk assessment models that capture all potential vulnerabilities, and in-depth research into ethical implications and governance frameworks for AI security. The substantial focus on identifying issues, particularly in Generative AI and large language models (LLMs) highlights a need to pivot toward developing practical solutions and robust safeguards for these evolving technologies. The call for a deeper emphasis on hardening AI systems and multimodal generative models reflects the importance of protecting these widely used systems from escalating security threats. Moreover, the existing challenges in AI security tools, such as adaptability, scalability, and transparency, also point to crucial areas for improvement. Addressing these limitations by enhancing tools to oper-

ate effectively in real-world conditions and expanding their scope to cover complex, multimodal models will significantly improve AI security and its practical applications across diverse sectors.

These gaps not only represent challenges but also opportunities for groundbreaking research that can significantly advance our understanding and capabilities in AI security. Looking ahead, it is absolutely imperative to adopt a proactive and multidisciplinary approach to AI security research as outlined in this review. This comprehensive approach should delve far beyond technical aspects and encompass the broader ethical, social, and legal implications of AI. Prioritizing practical real-world applications and establishing standardized benchmarks for AI security are crucial initial steps that will uphold the effectiveness and relevance of future research endeavors.

In conclusion, this systematic mapping study has not only collated and analyzed existing research in AI security but has also sought to chart a course for future investigations. By highlighting both the achievements and the gaps in current research, this study aims to inspire and guide future endeavors in AI security, ensuring that as AI technologies advance, they do so with robust security measures and ethical considerations at their core. It is through these continued efforts that we can hope to harness the full potential of AI while safeguarding against its inherent risks.

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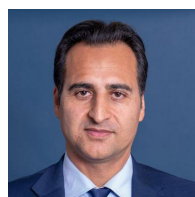
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